mbusgw-t2r

User Guide

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NaPi World & StarLet Squad collaboration

mbusgw-t2r User Guide

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Preface

1. About This Manual

This manual describes the configuration, the operation and the troubleshooting of mbusgw-t2r --- a gateway which translates MODBUS TCP requests coming from the network into MODBUS RTU transactions on a serial (RS-485 or RS-232) line, and returns the answers back.

2. Intended Audience

This manual is written for a reader with no prior MODBUS or Linux administration experience: every step is spelled out and nothing is assumed. An experienced engineer may go straight to Chapter 3 (the settings file) and Chapter 7 (the troubleshooting).

3. Document Structure

Chapter 1 explains what the gateway does. Chapter 2 is the pre-flight checklist. Chapter 3 describes the settings file key by key. Chapter 4 covers the start, the command line options and the first end-to-end test. Chapter 5 describes the built-in Time Stamp pseudo device. Chapter 6 teaches how to read the log. Chapter 7 is the troubleshooting reference: the symptom, the cause, the action, and the reference of all the message codes.

4. Related Documents

README.md --- the installation sequence (the StarLet package, the build, `make install`). MODBUS over Serial Line Specification and Implementation Guide and MODBUS Application Protocol Specification --- the protocol itself, available from modbus.org.

5. Conventions

Convention	Meaning
`Monospace type`	Code examples, file names, commands and interactive screen displays.
`%T2R-E-CODE`	A message code as it appears in the gateway log; see Section 7.2.
`<...>`	A placeholder to be replaced by an actual value, e.g. a device name.
`$`	The shell prompt of an unprivileged user; do not type it.
`Ctrl/C`	Hold down the key labeled Ctrl while you press the key C.

Chapter 1. What the Gateway Does

You have a device (an electricity meter, a temperature controller, a PLC ...) which talks MODBUS RTU over a serial line. And you have a program (a SCADA, a script, `mbpoll` ...) which talks MODBUS TCP over the network. The gateway sits between them and translates, as Figure 1 shows.

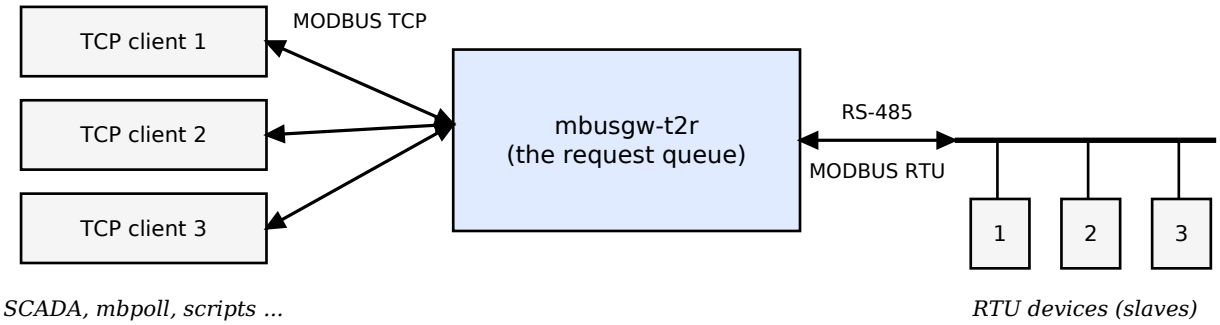


Figure 1. The data path of the gateway

Several TCP clients may connect at the same time: the gateway queues them so that only one request at a time goes to the serial line --- an RTU line physically cannot carry two transactions at once.

Several devices may share one RS-485 line. They are told apart by the slave address inside the MODBUS request itself; the gateway does not need to know their addresses at all.

Chapter 2. Before You Start

Walk this checklist once --- it saves an hour of guessing later.

1. The gateway is installed (see README.md). After the installation you have the program `/opt/mbusgw-t2r/sbin/mbusgw-t2r` and the settings file `/opt/mbusgw-t2r/etc/modbus-t2r_settings.conf`.

2. You know which serial port the device is connected to. On Linux it is a file like `/dev/ttyS0` or `/dev/ttyUSB0`. If unsure, plug the USB adapter out and in, then run:

```
$ dmesg | tail
```

3. You know the line parameters of the device: the speed, the data bits, the parity and the stop bits. They are in the device manual; the most common set is `9600, 8, N, 1`.

4. Your user may open the port. Check the group of the port file:

```
$ ls -l /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 0 ... /dev/ttyUSB0
```

If the group is `dialout`, add yourself to it and re-login (or simply run the gateway with `sudo`):

```
$ sudo usermod -a -G dialout $USER
```

Chapter 3. The Settings File

The settings file has two sections: `serials` (the serial lines) and `listeners` (the TCP ports). A minimal working example:

```
serials = (
{ device = "/dev/ttyUSB0";
  chars  = "9600, 8, N, 1";
  iotmo  = 1000;
}
);

listeners = (
{ bind   = "TCP:0.0.0.0:502";
  target = "/dev/ttyUSB0";
  connlm = 8;
}
);
```

Read it as: "open the port /dev/ttyUSB0 at 9600-8-N-1 and wait for an answer up to 1000 ms; listen for TCP clients on every interface, port 502, and forward them to that port".

3.1. The serials Section

One record per serial port. Table 3-1 lists every key.

Key	Required	Meaning and the allowed values
`device`	yes	The serial port file, e.g. /dev/ttyUSB0.
`chars`	yes	The line parameters: speed, data bits, parity, stop bits. Speed 50..4000000; data bits 5..8; parity N (none), E (even), O (odd); stop bits 1..2.
`iotmo`	no	How long to wait for the answer of the device, milliseconds. Default 1000.
`rs485`	no	1 --- ask the kernel to drive the RS-485 direction control (only for the ports which support it). Default 0.
`ts_enabled`	no	1 --- enable the Time Stamp pseudo device (Chapter 5). Default 0.
`ts_unit`	no	The slave address the pseudo device answers on. Default 135.
`ts_fncode`	no	The function code it answers to. Default 4.
`ts_base_reg0`	no	The first register of the time stamp. Default 135.

If a record is wrong, the gateway skips it and says why --- with the allowed range right in the message:

```
%T2R-E:  [serial #00:</dev/ttyUSB0>] --- speed 31 baud is out of range [50..4000000]
```

3.2. The listeners Section

Key	Required	Meaning and the allowed values
`bind`	yes	Where to listen: TCP:<IP address>:<port>. The port is 1..65535; 0.0.0.0 means every interface. UDP is not supported.
`target`	yes	Which serial port to forward to. Must match a device from serials exactly, character by character.

Key	Required	Meaning and the allowed values
`connlm`	no	How many TCP clients may wait in the connection queue, e.g. 8.
`iotmo`	no	The network I/O timeout, milliseconds.

Port 502 is the standard MODBUS TCP port, but on Linux the ports below 1024 need the root privileges. If you do not want to run as root, take e.g. `5502` and point your client there. Several listeners may share one target: the gateway serializes their requests.

Chapter 4. Running the Gateway

By hand, in a terminal (the messages go straight to the screen):

```
$ /opt/mbusgw-t2r/sbin/mbusgw-t2r /settings=/opt/mbusgw-t2r/etc/modbus-t2r_settings.conf
```

Option	Meaning
`/settings=<file>`	The settings file (Chapter 3).
`/trace`	The verbose tracing: every request and answer is dumped in hex. Priceless at the first run.
`/logfile=<file>`	Write the log to a file instead of the screen.
`/logsize=<octets>`	Rotate the log file above this size.

A healthy start prints (shortened):

```
%T2R-I-REVISNF, Rev: T2R X.00-06/aarch64(built at ...) (REV: 00.06.00)
%T2R-I:  Added device #00 [</dev/ttyUSB0>, Chars: <9600, 8, N, 1>, ...] --- added
%T2R-I:  Added listener #00 [Target: </dev/ttyUSB0>, Net: <TCP:0.0.0.0:502>, ...] --- added
%T2R-S-DEVREADY, Device </dev/ttyUSB0> [9600 baud, answer tmo: 1000 msec, t3.5: 4010 usec] --- is r
%T2R-S-LSNRRDY, [#3] Listener 0.0.0.0:502 [Target: </dev/ttyUSB0>] --- is ready
```

Two lines matter most: DEVREADY (the port is open) and LSNRRDY (the TCP port is listening). If you see both, the gateway is up. Figure 2 shows what happens to every request after that.

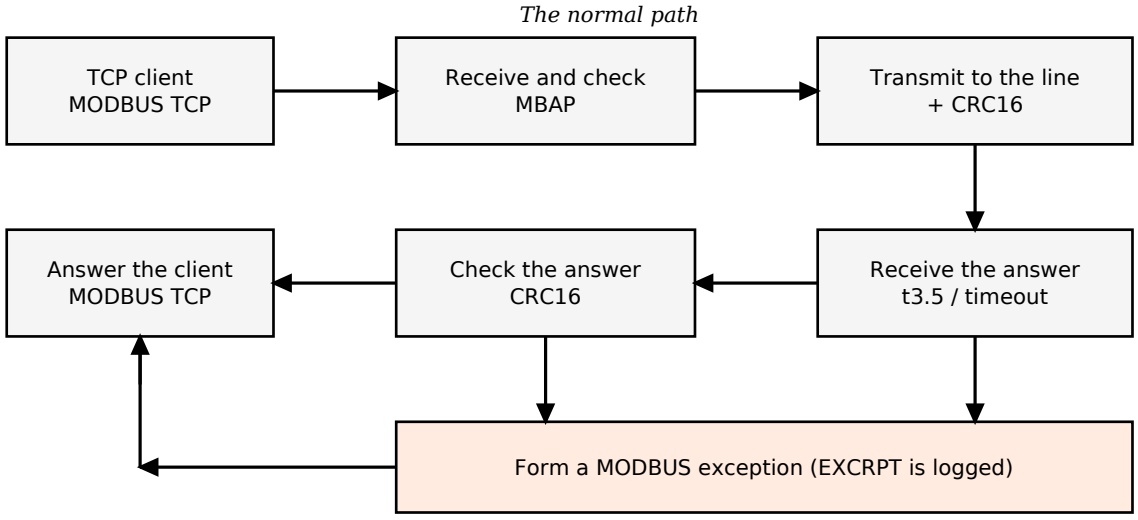


Figure 2. The lifecycle of one request

To stop the gateway press Ctrl/C once and wait a second. To toggle the tracing of a running gateway without a restart:

```
$ kill -USR1 <pid>
```

4.1. The First Test with mbpoll

`mbpoll` is a small free MODBUS client (`sudo apt install mbpoll`). Read 3 holding registers from the device with the slave address 1:

```
$ mbpoll -a 1 -r 1 -c 3 -1 127.0.0.1 -p 502
```

If the numbers come back --- everything works, end to end.

Chapter 5. The Time Stamp Pseudo Device

The gateway can answer by itself --- without touching the line --- to a read request sent to a special slave address. The answer is the current time of the gateway host. This lets a SCADA read the precise time over the plain MODBUS.

With `ts_enabled = 1; ts_unit = 135; ts_fncode = 4; ts_base_reg0 = 135;` a request "read N input registers starting at 135 from the slave 135" returns the registers of Figure 3.

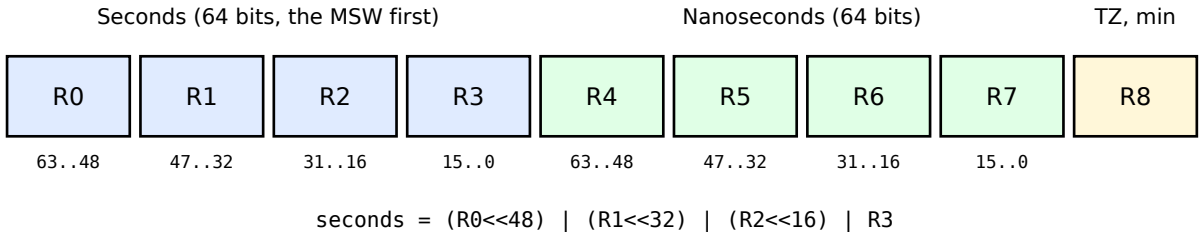


Figure 3. The Time Stamp registers layout

Registers requested	You get
4	The seconds of the UNIX epoch, 64 bits, the most significant 16-bit word first.
8	Plus the nanoseconds, packed the same way into R4..R7.
9	Plus R8 = the local timezone offset, in signed minutes (e.g. Moscow = +180).

Any other register count in a TS request is answered with the MODBUS exception ILLEGAL_DATA_ADDRESS --- this is on purpose.

Chapter 6. Reading the Log

Every important event is one line with a code:

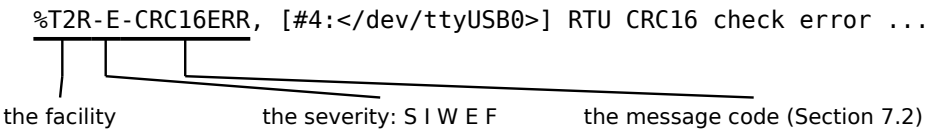


Figure 4. The anatomy of a log message

The severity letter is: S --- success, I --- information, W --- warning, E --- error, F --- fatal. The part `[#4:</dev/ttyUSB0>]` means: the file descriptor #4, this serial port.

Grep the log by the code --- this is exactly why the codes exist:

```
$ grep CRC16ERR /var/log/mbusgw-t2r.log
```

Chapter 7. Troubleshooting

The golden rule: run with /trace and read the log. The gateway always says what it dislikes.

7.1. The Symptom, the Cause, the Action

You see	It means / what to do
`%T2R-E-DEVOPNERR, ... errno: 2`	The port file does not exist: the adapter is unplugged or the name is wrong. Run dmesg | tail after plugging; fix device in the settings.
`%T2R-E-DEVOPNERR, ... errno: 13`	Permission denied. Run with sudo, or add yourself to the dialout group.
`%T2R-E-DEVOPNERR, ... errno: 16`	The port is busy. Find the holder: sudo fuser /dev/ttyUSB0, and stop it.
`%T2R-E-NOANSWER, ...`	The request went out, nothing came back: 1) a wrong slave address in the client request; 2) the wiring (swap A/B of RS-485); 3) a wrong speed/parity --- re-check chars; 4) the device is off.
`%T2R-E-CRC16ERR, ...`	The octets arrive damaged: 1) chars does not match the device (the most common!); 2) the line noise --- the grounding, the termination, the cable length; 3) two masters on one line.
`%T2R-E-BADFRAME, ...`	Something arrived, but it is not a sane MODBUS frame: usually the same causes as CRC16ERR; 400+ octets of garbage mean a foreign, non-MODBUS device on the line.
`CRC16ERR / BADFRAME now and then, on an FTDI USB adapter only`	The adapter gives the answer by pieces, once in 16 msec (latency_timer). Since X.00-10 the gateway waits for the whole answer of the functions 01-06, 15, 16, 23; the others are still cut. Set latency_timer to 1 msec (the answers get faster too): copy /opt/mbusgw-t2r/share/doc/mbusgw-t2r/99-ftdi-latency.rules to /etc/udev/rules.d/, run sudo udevadm control --reload && sudo udevadm trigger --subsystem-match=usb-serial; cat /sys/bus/usb-serial/devices/ttyUSB0/latency_timer must print 1.
`%T2R-W-FRAMETMO, ...`	The answer started but never finished: a very slow or hanging device. Raise iotmo; check the cable.
`%T2R-W-EXCRPT, ... (SERVER_DEVICE_FAILURE)`	The gateway told the client "the serial side failed". The real reason is in the previous log line (NOANSWER, CRC16ERR ...).
`%T2R-E-LSNRERR, ... errno: 98`	The TCP port is taken: a second gateway instance, or another program. Find it: sudo ss -tlnp | grep 502.
`%T2R-E-LSNRERR, ... errno: 13`	The ports below 1024 need root. Run with sudo, or take a port above 1024 (e.g. 5502).
`... out of range [a..b]`	A settings value is out of its range; the allowed range is right in the message. Fix the value.
`No serials has been defined!`	Not a single serials record survived the validation. Read the error lines above --- each rejected record says why.
`The data is wrong or shifted`	The byte order confusion on the client side: the MODBUS registers are the big-endian 16-bit words; check how the client assembles the 32/64-bit values.

7.2. The Message Codes Reference

Code	Sev	When it appears
`REVISNF`	I	At the start: the program version. Quote it when asking for help.
`DEVREADY`	S	The serial port is open; the actual baud, the answer timeout and the t3.5 interval are shown.
`LSNRRDY`	S	The TCP port is listening; the address, the port and the target device are shown.
`NETCONN`	S	A TCP client has connected; its address:port and the listener are shown.
`NETDISCN`	S	A TCP client has disconnected; its address:port are shown.
`DEVOPNERR`	E	The serial port cannot be opened; errno says why (2 = no such file, 13 = the permissions, 16 = busy).
`LSNRERR`	E	bind()/listen() failed for a TCP port; errno says why (98 = taken, 13 = the privileges).
`NOANSWER`	E	The device did not answer within the iotmo timeout.
`FRAMETMO`	W	The answer began but was not completed within the timeout.
`BADFRAME`	E	The received frame length is outside the sane [5..256] octets range.
`CRC16ERR`	E	The checksum of the received frame does not match; both values are shown.
`EXCRPT`	W	A MODBUS exception is returned to the TCP client; the code and its name are shown.
`EXITST`	I	The gateway exits; the exit flag and the final status are shown.

7.3. If Nothing Helps

Collect and attach to your question: 1) the full start-up log with /trace (from REVISNF to the first error); 2) your settings file; 3) the output of `ls -l <device>` and `dmesg | tail -20`; 4) the exact model of the RTU device and its documented line parameters. With these four things the problem is almost always visible at a glance.